

Matheus Hoffmann



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me

sr. software engineer with **8 years** of experience building scalable (and occasionally clever) solutions mainly for payment processing & fraud-fighting, now building open-source ocean tech at **marola-dev**

open source — marola-dev

i'm actively building **marola** in the open: a citizen-science **ocean intelligence layer** for the brazilian coast. it answers “*can i swim tomorrow, and when?*” for every beach around florianópolis, rio de janeiro and salvador: the best hour from swell, wind, tide, uv and the official bathing-water bulletins, with the *why* in plain language, live at **marola.dev**.

- water quality first** - every official sampling point (ima/sc, inea, inema) is on the map, coloured by its last bulletin; unfit water zeroes the score in deterministic scala 3 (kyo) code, and the llm explains but never overrules
- the sea in front of you** - tide, swell, wind, uv, jellyfish and whale odds per beach, a sourced *did you know?* about the local sea, and *ask the ocean*: rag with citations on ollama, an mcp tool server and its own small model, marola-sea, no account or api key
- designed in public** - every non-trivial change starts as a public design doc (MIPs); next up: wind, wave, satellite and sea-heat map layers, *ressaca* (storm surf) warnings, wave-model ensembles with WAVEWATCH III and an open data lake of beaches and water quality

[marola.dev](#) FOSS [marola-dev](#) docs [docs.marola.dev](#) license MIT DOI [10.5281/zenodo.23224155](#)
[civic-tech](#) [disaster-risk-management](#) [forecasting](#) [meteorology](#) [non-profit](#) [oceanography](#) [water-quality](#)

🔒 contributors welcome, humans and ai agents alike — no code needed: use the map and tell us where it's wrong, share local sea knowledge, or open an issue or a PR at **marola-dev/marola**. or mail me: **mhoffmannfs[at]gmail.com**

research — ww3-gpu

my undergraduate project at ufrj (escola politécnica), co-advised at labeco/ufsc: **ww3 gpu lab**, an open lab for running noaa's **WAVEWATCH III** wave model and moving its expensive kernels to gpus *without changing the answer*. a nix-pinned fortran/mpi/netcdf toolchain builds ww3 and runs a regression test in one command. the dia nonlinear-interaction kernel (W3SNL1), ported to c++/**kokkos**, matches the fortran bit for bit on serial, openmp and cuda, and runs 1,000 sea points in 0.047 ms on an rtx 4090 against 24.96 ms serial. it also holds a 16-lesson course, from the wave spectrum to porting kernels with coding agents.

research [ww3-gpu](#) DOI [10.5281/zenodo.23221351](#)

exp-highlights

- itv (uk)** - ml/mlops on aws for channel audience forecasting (2025/aug 2026)
- signifyd (us - award winner tenacious feb/2024)** - ml eng. for loss forecasting & risk-analysis on databricks (2023/2024)
- elemeno AI (us - br client bvmf: AMER3)** - mlops engineer with k8s on gcp (2021/2022)
- broad (uk - YC W21)** - sr. software engineer with scala for digital banking on gcp (2020)
- stone (br: nasdaq: STNE)** - ml & software engineer with scala on aws (2017-2019)
- LIOC (br: UFRJ/COPPE)** - ocean instrumentation lab - scientific research, ROV, microcontrollers (2016/2017)

certs

AWS [Cloud Practitioner](#) AWS [ML Specialty](#) AWS [ML Engineer Associate](#) AWS [Data Engineer Associate](#)

learning

GCP [Professional Agentic Architect \(studying\)](#)

misc

Strava [open water swimming](#)
PADI [Advanced Open Water Diver](#) PADI [Enriched Air Nitrox Diver](#) PADI [Rescue Diver](#)